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Structure of the photosynthetic Calvin-Benson-Bassham sedoheptulose-1,7-bisphosphatase SBPase from the model microalga *Chlamydomonas reinhardtii*

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Abstract

The Calvin-Benson-Bassham cycle (CBBC) performs carbon fixation in photosynthetic organisms. Among the eleven enzymes that participate in the pathway, sedoheptulose-1,7-bisphosphatase (SBPase) is expressed in photo-autotrophs and contributes to the regeneration of ribulose-1,5-bisphosphate, the carbon fixation co-substrate used by ribulose-1,5-bisphosphate carboxylase/oxygenase (Rubisco). While SBPase is structurally similar to fructose-1,6-bisphosphatase (FBPase) involved in both neoglucogenesis and the CBBC, it exclusively functions in the CBBC and is indispensable for building a productive cycle. In this study we report the first structure of an SBPase from a chlorophyte, the model unicellular green microalga *Chlamydomonas reinhardtii*. By combining experimental and computational structural analyses, we describe the topology, conformations and quaternary structure of *Chlamydomonas reinhardtii* SBPase. We identify active site residues and locate sites of redox- and phospho-post-translational modifications that contribute to enzymatic functions. Finally, we observe that CrSBPase adopts distinct oligomeric states that may dynamically contribute to the control of its activity.

eLife assessment

The manuscript reports **useful** findings by resolving the crystal structure of Sedoheptulose-1,7-Bisphosphatase (SBPase) from the green algae *Chlamydomonas reinhardtii*, which is involved in the Calvin cycle. The data presented are **solid** based on validated methodologies, which help in understanding the structure and function of this enzyme.

Introduction

Photosynthetic carbon fixation is performed by eleven enzymes that collectively operate the Calvin-Benson-Bassham cycle (CBBC) (Calvin 1962; Johnson 2016), wherein Rubisco plays the prominent role in catalyzing fixation of inorganic carbon onto a pentose acceptor: ribulose-1,5-bisphosphate (RuBP). The net gain of the CBBC is one triose in the form of glyceraldehyde-3-phosphate (G3P) for every three Rubisco catalyzed carboxylation reactions (Johnson and Alric 2013). Regeneration of RuBP is ensured by the other ten enzymes of the CBBC, among which phosphoribulokinase (PRK) catalyzes the phosphorylation of ribulose-5-phosphate (R5P) and sedoheptulose-1,7-bisphosphatase (SBPase) catalyzes the hydrolysis of sedoheptulose-1,7-bisphosphate (SBP) into sedoheptulose-7-phosphate (S7P). Remaining CBBC enzymes are paralogs of those involved in cytoplasmic glycolysis, neoglucogenesis or the pentose-phosphate pathway. Hence, Rubisco, PRK, and SBPase are expressed in photosynthetic organisms and fulfil biochemical functions so far uniquely attributed to the photosynthetic CBBC. The CBBC is perfectly conserved in plants and cyanobacteria since its enzymes are expressed in oxygenic prokaryotes and plant chloroplasts, which consist in subcellular compartments that evolved from ancestral endosymbiosis between cyanobacteria and heterotrophic eukaryotic cells. For instance, the analysis of PRK from *Synechococcus* PCC 6301, *Arabidopsis thaliana*, and *Chlamydomonas reinhardtii* display high structural similarities (Gurrieri, Del Giudice et al. 2019; Wilson, Hayer-Hartl et al. 2019).

Native SBPase was purified from wheat chloroplasts (Woodrow 1982) and corn leaves (Nishizawa and Buchanan 1981), allowing quantitative assays of its catalytic function and the *in vitro* confirmation of magnesium-dependent activation and activity modulation by pH and redox switch mechanisms (Anderson 1974; Woodrow and Walker 1980; Cadet and Meunier 1988). Redox control is exerted through a dithiol/disulfide interchange dependent upon the ferredoxin-thioredoxin cascade arising from the activity of the illuminated photosystems (Schürmann and Buchanan 1975). In tobacco leaves extracts, SBPase reaches its maximal activity after 10 minutes illumination (Zimmer, Swart et al. 2021). This delayed activation kinetics is consistent with the slow reduction pattern of *Arabidopsis* SBPase *in vitro* (Yoshida and Hisabori 2018).

The crystal structure of SBPase from the moss *Physcomitrium patens* (PpSBPase) has been solved at a resolution of 1.3 Å (Protein Data Bank entry : 5iz3) (Gutle, Roret et al. 2016). PpSBPase fold is highly similar to that of moss FBPase, yet the protein was shown to distinctively oligomerize as a homodimer and to form a disulfide bridge in a different motif than that previously described for FBPase (Chiadmi, Navaza et al. 1999). In spite of their homology, the sequences of FBPase and SBPase seem to have evolved different regulatory properties that allow different contributions to the CBBC. Recently, it was proposed that the CBBC can bypass fructose-1,6-bisphosphate (FBP) altogether, by only relying on SBP, glycolysis and the oxidative pentose-phosphate pathway paralogs to regenerate the substrate of Rubisco (Ohta 2022). Engineered SBPase overexpression enhances photosynthesis and growth in plants (Lefebvre, Lawson et al. 2005; Feng, Han et al. 2007; Feng, Wang et al. 2007; Rosenthal, Locke et al. 2011; Simkin, Lopez-Calcano et al. 2017) and in the green microalga *Chlamydomonas reinhardtii* (Hammel, Sommer et al. 2020), indicating that SBPase appears as a metabolic bottleneck for the CBBC along with Rubisco. Based on these evidences, it is plausible to consider that SBPase is a good target to improve the photosynthetic capacity by means of metabolic engineering coupled to synthetic biology (Kubis and Bar-Even 2019), especially in the experimentally convenient multi-omic model alga *Chlamydomonas* (Mettler, Muhlhaus et al. 2014; Schmollinger, Muhlhaus et al. 2014).

In *Chlamydomonas*, SBPase is encoded by a single nuclear gene (Cre03.g185550), and the protein (Uniprot entry P46284) is addressed to the chloroplast by a 54-residue amino-

terminal peptide (Emanuelsson, Brunak et al. 2007; Almagro Armenteros, Salvatore et al. 2019). SBPase abundance in the algal cell was quantified by mass spectrometry as $1.1\text{--}1.3 \pm 0.3$ amol/cell, representing 0.15 % of total cell proteins (Hammel, Sommer et al. 2020). In *Chlamydomonas* protein extracts, SBPase was shown to undergo post-translational modifications by phosphorylation of serines and threonines (Werth, McConnell et al. 2017), and redox modifications of cysteine thiols (Zaffagnini, Bedhomme et al. 2012; Morisse, Zaffagnini et al. 2014; Pérez-Pérez, Mauriès et al. 2017). However, the mechanisms by which post-translational modifications affect the catalytic activity of CrSBPase is still lacking. Here, we describe the crystal structures of CrSBPase in two redox states, assay its activity *in vitro*, and investigate the correlation between the redox state and the oligomerization of the protein. Our results suggest that a redox control over oligomer equilibrium could represent a novel mechanism to regulate enzymatic activities in the CBBC.

Material and methods

Chemical reagents were purchased from Merck (Darmstadt, Germany), chromatography material was provided by Cytiva (Vélizy-Villacoublay, France), crystallization solutions and consumable were purchased from Qiagen (Hilden, Germany), Hampton Research (Aliso Viejo, CA USA), and SPT Labtech (Melbourn, UK).

Cloning and mutagenesis of CrSBPase expression plasmids

The sequence coding for full-length mature SBPase was amplified by reverse transcriptase polymerase chain reaction (RT-PCR) from *Chlamydomonas reinhardtii* total ribonucleic acids extract and inserted at sites 5'-NdeI and BamHI-3' into expression vector pET3c modified for the fusion of an amino-terminal hexa-histidine tag in the recombinant protein, yielding plasmid pSL-175-His₆-CrSBPase-WT. PCR primers were 5'-TCTCGCCATATGGCCGTCTGACCCAGGCC-3' (forward, NdeI site underlined) and 5'-CGGGTGGGATCCTTAGGCAGCCACCTTCTC-3' (reverse, BamHI site underlined). The encoded protein encompasses 331 amino acids, with a molar mass of 36,176.7 Da and a theoretical extinction coefficient of $24,722.5 \text{ M}^{-1} \cdot \text{cm}^{-1}$. Unless otherwise stated, amino acid residues numbering is that of GenBank entry DAA79954.1. Point mutants of cysteines 115 and 120 singly substituted with serines were generated by site-directed PCR mutagenesis with primers listed in [supplementary table 1](#), producing plasmids pSL-181-His₆-CrSBPase-C115S, and pSL-182-His₆-CrSBPase-C120S.

Recombinant CrSBPase purification

Plasmids pSL-175, -181, and -182 were used to transform *Escherichia coli* BL21(DE3). The transformants were grown at 37 °C in 2YT supplemented with 150 µg/mL ampicillin. When cultures reached an OD₆₀₀ of ~0.6, T7-dependent CrSBPase overexpression was induced by the addition of isopropyl-β-D-thiogalactopyranoside (IPTG) at 0.2 mM for 3 h.

Harvested cell pellets were resuspended in 20 mmol.L⁻¹ Tris pH 7.9, 100 mmol.L⁻¹ NaCl (buffer A) and lysed by 1 sec / 1 sec pulsed sonication for 3 min. The soluble fraction of the lysate was separated by 20 min centrifugation at 20,000 g and loaded on 2 mL Ni-NTA resin for His-tag mediated affinity chromatography. The resin was washed in four steps with 25 mL buffer A, 25 mL buffer A supplemented with 10, 20, and 30 mmol.L⁻¹ imidazole, and CrSBPase was eluted with 10 mL buffer A supplemented with 100, 200, and 300 mmol.L⁻¹ imidazole. Pooled eluates containing CrSBPase were loaded on HiLoad Superdex 200 26/600 size-exclusion chromatography column and eluted in buffer A. Protein purity was assessed

by electrophoresis on 12% acrylamide gel under denaturing and reducing conditions (supplementary figure 1). Peak fractions were assembled and concentrated by ultrafiltration on Amicon units of MWCO 30,000 Da cut-off. Final concentrations of purified proteins were in the 1-10 mg.mL⁻¹ range as measured by Nanodrop ND-2000 spectrophotometer.

Western blot

Polyclonal antibodies were generated by immunization of rabbits with pure recombinant CrSBPase (Genecust, Boynes France). Analyzed fractions were separated by denaturing polyacrylamide electrophoresis and transferred to nitrocellulose membrane for detection with primary antibody subsequently detected by secondary anti-rabbit antibodies coupled to horseradish peroxidase. Detection was done with commercial ECL peroxidase assay (GE Healthcare, Chicago IL USA) with a Chemidoc (Bio-Rad, Hercules CA USA).

Crystallization and structure determination

Pure recombinant His6-CrSBPase-WT concentrated at 3-6 mg/mL was tested for crystallization on screens JCSG I-IV in 200 nL sitting drops. Single crystals were obtained and optimized with mother liquor 0.1 mol.L⁻¹ sodium HEPES pH=7.5, 2% (v/v) polyethylene glycol 400, 2.0 mol.L⁻¹ ammonium sulfate, flash-frozen in mother liquor supplemented with 25% glycerol and tested for X-ray diffraction. Complete X-ray diffraction datasets were collected at SOLEIL beamline Proxima-2A and allowed the determination of the crystal structure by model replacement with 5iz3, the structure of ortholog SBPase from the moss *Physcomitrium patens*. Model building in COOT and refinement in the PHENIX software suite yielded structure 7b2o (Emsley, Lohkamp et al. 2010) (Adams, Afonine et al. 2011) (Liebschner, Afonine et al. 2019). After we obtained our experimental structure, the European Bioinformatics Institute (Hinxton, UK) communicated high-accuracy prediction models computed by Deepmind ALPHAFOLD2 (Jumper, Evans et al. 2021), including a prediction of CrSBPase structure (<https://alphafold.ebi.ac.uk/entry/P46284>) that matches 7b2o crystal structure with RMSD=0.453 Å.

Hence, X-ray crystallography and ALPHAFOLD independently contribute to the proposed structural analysis. In order to determine the structure of CrSBPase in a reduced state, crystals were grown in the presence of 10 mmol.L⁻¹ of the reducing agent tris-(2-carboxyethyl)phosphine (TCEP) in condition JCSG IV E8/56 0.2 M lithium sulfate, 0.1 M Tris pH 7.0, 1.0 M sodium/potassium tartrate. Crystals were cryo-protected in mother liquor supplemented with 25% ethylene glycol, flash-frozen and tested for X-ray diffraction at SOLEIL beamline Proxima-2A. A complete dataset was collected that allowed the determination of the structure by molecular replacement with 7b2o as a template. Among the eight CrSBPase copies of the asymmetric unit, two had their disulfides absent as a consequence of the TCEP treatment while the others six subunits were essentially identical to untreated structures in entry 7b2o.

Enzymatic assays

The catalytic activity of 10-200 nmol.L⁻¹ pure recombinant CrSBPase was spectrophotometrically assayed in buffer A by coupling the hydrolysis of 1 M fructose-1,6-bisphosphate (FBP) into fructose-6-phosphate (F6P) to the reduction of 0.5 M NADP⁺ into NADPH through isomerization of F6P into glucose-6-phosphate (G6P) by 0.5 U.mL⁻¹ phosphoglucose isomerase (PGI) and the oxidation of G6P into 6-phosphogluconolactone (6PGL) by 0.1-0.5 U.mL⁻¹ glucose-6-phosphate dehydrogenase (G6PDH). Reporter assay is coupled in a molar ratio FBP:F6P:G6P:6PGL of 1:1:1:1 to the oxidation of one molar equivalent of NADP⁺ into NADPH that we recorded over 0-20 minutes by measuring the

absorbance at 340 nm using a UVIKON spectrophotometer. Pre-treatment with 0-100 mmol.L⁻¹ DTT, 0-30 mmol.L⁻¹ MgSO₄ and 0-15 μmol.L⁻¹ recombinant CrTRX-f2 supplemented with 1 mM DTT (Lemaire, Tedesco et al. 2018) was carried out for 30 min at room temperature before adding the enzyme in the assay solution.

Titration of accessible cysteine thiols

Quantification of free reactive cysteines on recombinant CrSBPase was performed with 5,5'-dithio-bis-(2-nitrobenzoic acid) (DTNB, Ellman's reagent) mixed with the purified recombinant protein in a 200:1 molar ratio in a 1 mL reaction volume containing 100 mM Tris-HCl, pH 7.9. Reaction of DTNB with solvent-exposed cysteine thiols produces 2-nitro-5-thiobenzoate (TNB⁻) that we quantified by monitoring the absorbance at 412 nm and a molar extinction coefficient of TNB⁻ of 14.15 mM⁻¹.cm⁻¹. We calculated the number of free reactive cysteines per CrSBPase monomer by dividing the molar concentration of TNB⁻ by that of the protein at equilibrium (t >60 min).

Molecular dynamics

Molecular dynamics (MD) simulations were performed with the Gromacs package (v2021.4). A dimer of CrSBPase was solvated in a rhombic dodecahedron box with at least 8 Å between the protein atoms and the box edges, which corresponds to ~64000 atoms in total (~18000 water molecules) and a triclinic box of 97×97×68 Å³. The box was neutralized with 18 sodium atoms to make it neutral. The starting conformation was the crystallized structure of oxidized CrSBPase, and we protonated C115 and C120 to manually create the reduced state. The protein was described with the Amber14SB force field and water with the TIP3P model. Non-bonding interactions were described with standard parameters: van der Waals interactions had a cut-off of 8 Å, and electrostatic interactions were computed with PME with default parameters with a separation between spaces at 8 Å. Bonds containing a hydrogen atom were restrained with the LINCS algorithm with default parameters. After an energy minimization, the system was equilibrated under the NPT conditions with a δt=1 fs timestep, the velocity-rescale thermostat and the Berendsen barostat. Velocities were generated at 100 K and the system was heated up to 300 K in 400 ps before performing 100 ps at 300 K. We then ran simulations of 20 μs under the NPT conditions with a δt=2 fs timestep, the velocity-rescale thermostat and the Parrinello-Rahman barostat.

Small-angle X-ray scattering

Pure recombinant CrSBPase was injected on Agilent HPLC BioSEC-3 300 column in buffer A, in line with the X-ray beam capillary at SOLEIL beamline SWING. CrSBPase was untreated or pre-treated with 10 mmol.L⁻¹ of reducing agent dithiothreitol (DTTred) or oxidizing agent trans-1,2-dithiane-4,5-diol (DTTox). Diffusion curves were analyzed with ATSAS software to determine the molecular mass and oligomeric state of the protein.

Results

Crystal structure of CrSBPase

Pure recombinant CrSBPase crystallized in space group P2₁2₁2 and provided a complete X-ray diffraction dataset at a resolution of 3.09 Å. The asymmetric unit is composed of six polypeptide chains packing as dimers of dimers (see below) and four water molecules. All chains align within a root mean square deviation (RMSD) <0.25 Å and can be considered

identical. Three hundred and nine out of the 331 residues were built in the electron density, the unmodelled sequences are in the disordered extensions at the amino- and carboxy-termini of the model. CrSBPase folds into two distinct domains (figure 1A-B), an amino-terminal domain composed of a mixed β -sheet formed by strands 3-9 and a carboxy-terminal domain composed of a mixed β -sheet formed by strands 10-13. α -helices 1-3 pack on the amino-terminal end of the seven-strand sheet, while α -helices 4,5, and 8 pack on the carboxy-terminal end of the four-strand sheet. α -helices 6 and 7 are sandwiched in between the two sheets. A short antiparallel β -sheet is constituted of 4-residue strands 1 and 2 separated by a β -turn and preceded by loop A₁₁₃SCAGTAC₁₂₀. This motif projects to solvent from α -helices 2 and 3. Residues 113-130 forming the loop and β -hairpin (LBH) motif are positioned away from the core of the protein and will be further discussed in the next sections.

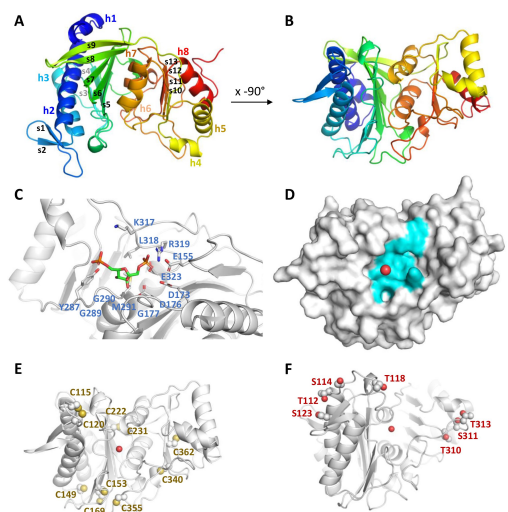


Figure 1.

Crystal structure of CrSBPase.

(A) Topology is displayed as cartoon with main chain coloured from blue (amino-terminus) to red (carboxy-terminus). (B) Rotated view of A. by x-axis over -90° . (C) Putative active site residues inferred from alignment with FBPase bound to FBP (5l0a, human muscle fructose-1,6-bisphosphatase E69Q mutant in active R-state in complex with fructose-1,6-bisphosphate) are represented in sticks. (D) Surface representation in the B. orientation with putative active site residues coloured in cyan. Water molecule H₂O 401 oxygen is represented as a red sphere. (E) Cysteines site chains are represented in spheres. (F) Threonines and serines side chains reported to be the target of phosphorylations are represented in spheres.

CrSBPase catalytic pocket

CrSBPase is structurally similar to its ortholog from the moss *Physcomitrium patens* (Gutle, Roret et al. 2016). SBPase is a homolog of FBPase, another enzyme involved in the CBBC in photosynthetic organisms and also ubiquitously involved in neoglucogenesis. Cytosolic isoforms of FBPase from non-plant sources have been extensively studied (Moorhead, Hodgson et al. 1994; Lee and Hahn 2003). For instance, the crystal structure of human muscle FBPase complexed with its substrate (FBP) was solved (PDB entry: 5l0a, to be published). We aligned the structure CrSBPase onto that of human FBPase:FBP complex, confirming their similarity (RMSD = 0.824 Å) with the notable exception of the LBH motif which is absent in the human enzyme. In a 4-Å distance to FBPase co-crystallized with FBP we aligned 12 residues that are likely available for FBP or SBP binding by CrSBPase (figure 1C): E155, D173, D176, G177, Y287, G289, G290, M291, K317, L318, R319, and E323. These residues are exposed to the solvent in a continuous surface (figure 1D). Water molecule 401 could be modelled at the centre of the putative SBPase catalytic pocket, in the vicinity of E155, D173, D176, R319, and E323, in the catalytic pocket predicted from alignment with human FBPase bound to FBP. W401 specifically is in bonding distance to catalytic residues D178, D181, and E328. W401 is likely to react with the substrate SBP during hydrolysis.

Sites of redox post-translational modifications

Ten cysteines could be located in the model. Previous redox-based proteomic studies established that SBPase cysteines 115, 120, 222, 231, 355, and 362 are redox-modified by thioredoxin-dependent dithiol/disulfide exchange ((Lemaire, Guillon et al. 2004), Pérez-Pérez, Mauriès et al. 2017), S-glutathionylation on unidentified cysteines (Zaffagnini, Bedhomme et al. 2012), or S-nitrosylation at C355 and C362 (Morisse, Zaffagnini et al. 2014). C222, C231, C355, and C362 are putative targets of thioredoxin. DTNB-based thiol titration on the native purified SBPase indicates that 4.0 ± 0.6 cysteines are likely reactive owing to their relative exposure to solvent in a free thiol form. Amino acid residue accessible surface area and accessibility (ASA) calculation is consistent with DTNB titration, with C149, C153, C231, and C355 exposed to the solvent (ASA respective areas: $0.154 \text{ \AA}^2$, $0.113 \text{ \AA}^2$, $0.308 \text{ \AA}^2$, and $10.083 \text{ \AA}^2$), and the six others either shielded from solvent (C169, C222, C340, and C362) or engaged in a disulfide bridge (C115, C120) (Ahmad, Gromiha et al. 2004). In CrSBPase crystal structure, C222 and C231 are positioned on neighbouring strands of the amino-terminal domain, exposing their thiol $3.5 \text{ \AA}$ away from each other (Figure 1E). The formation of an intramolecular disulfide bridge at this cysteines pair is possible if local main chain rearrangements allow a closer contact. C362 thiol is positioned $10.5 \text{ \AA}$ away from the nearest thiol group of C340. Additionally, C340 and C362 are exposed on an opposite side of the carboxy-terminal β -sheet, disfavoring the formation of an internal disulfide bridge. C355 points its side chain towards C153 and C169 with a thiol-thiol distance of $4.6 \text{ \AA}$ and $10.1 \text{ \AA}$, respectively. Although not observed in our model, a C355-C153/C169 dithiol-disulfide exchange is thus plausible, as described for plant FBPase (Jacquot, Lopez-Jaramillo et al. 1997). Local rearrangements that occur upon FBPase disulfide reduction causes a larger scale relaxation of the active site required to reach maximal catalytic potential (Chiadmi, Navaza et al. 1999) as kinetically characterized for *Arabidopsis* (Yoshida and Hisabori 2018) and *Nicotiana* enzymes (Zimmer, Swart et al. 2021). In SBPase, the FBPase-type regulatory disulfide is not observed but the C115-C120 bridge in the LBH motif may regulate the enzyme by alternative conformational rearrangements. Indeed, the disulfide-bonded cysteines constrain loop A₁₁₃SCAGTAC₁₂₀ into a compact lasso conformation that could open into a relaxed conformation hypothetically activating catalysis.

Sites of post-translational phosphorylations

CrSBPase was reported to be phosphorylated from algae extracts at residues T112, S114, T118, S123, T310, S311, and T313 (Wagner, Gessner et al. 2006; McConnell, Werth et al. 2018). Consistently with the proximity of these residues in two sectors of the sequence (supplementary figure 2), the candidate modification sites are located in two contiguous regions in the three-dimensional model (figure 1F). T112, S114, T118, and S123 are exposed to the solvent of the amino-terminal domain. The residues belong to or are located nearby the A₁₁₃SCAGTAC₁₂₀ motif where phosphorylations may cross-signal with redox modifications on C115 and C120. T310, S311, and T313 are located on the other edge of the enzyme, at the solvent exposed tip of the carboxy-terminal domain. Their co-localization at the T₃₁₀SPT₃₁₃ motif likely facilitates a coordinated phosphorylation of the three residues by the same kinase. Because both groups of potential phospho-sites are located at $17\text{--}18 \text{ \AA}$ from the catalytic pocket (H₂O 401 taken as a reference point), there is no straightforward mechanism by which such modifications would exert a control over CrSBPase activity. Analysis of phospho-mimicking mutants along with the identification of the actual kinases/phosphatases couples that control SBPase phosphorylation state would allow to comprehend the role of this putative regulatory mechanism.

CrSBPase enzymatic activity

Recombinant CrSBPase was assayed for its capacity to catalyze FBP hydrolysis into F6P (Gutle, Roret et al. 2016), a proxy for the hydrolysis of SBP into S7P that we could not test by lack of available substrate and kinetic reporter method. Previous studies demonstrated that plant SBPase requires the cofactor Mg^{2+} and a chemical reductant (DTT) to be fully activated *in vitro* (Anderson 1974; Woodrow and Walker 1980; Cadet and Meunier 1988). In agreement with this, we found a total absence of activity in the untreated CrSBPase (data not shown), whereas pre-incubation of the enzyme with DTT and Mg^{2+} resulted in a strong activation (figure 2). By testing different concentrations of $MgSO_4$ (3, 10, 15, and 20 mM) and DTT (1, 10, 25, 50, and 100 mM), we established that 10 mM of DTT and 10 mM $MgSO_4$ are the optimal condition to attain the maximum specific activity which corresponds to $12.2 \pm 2.2 \mu\text{mol(NADPH) min}^{-1} \text{mg(SBPase)}^{-1}$ (figure 2A and 2B). Unexpectedly, DTT concentrations above 10 mM had an inhibitory effect, causing a drastic drop in activity that probably resulted from structural effects leading to denaturation or aggregation of the enzyme (figure 2A). In contrast, we observed no such inhibitory effect in the presence of magnesium at concentrations above 10 mM (figure 2B). Under physiological conditions, the light-dependent redox regulation of plant SBPase is specifically exerted by f-class thioredoxins (Gutle, Roret et al. 2016), which are reduced by the ferredoxin-thioredoxin cascade (Schürmann and Buchanan 1975). The chloroplast TRX-f2 from *Chlamydomonas* was thus tested and validated for its capacity to activate CrSBPase by reduction (figure 2C). The maximum activity, corresponding to $12.2 \pm 2.2 \mu\text{mol(NADPH) min}^{-1} \text{mg(SBPase)}^{-1}$, was obtained by pre-incubating the enzyme with 1 μM CrTRX-f2 supplemented with 1 mM DTT and 10 mM $MgSO_4$, and maintained even at higher CrTRX-f2 concentrations (i.e., 5 and 10 μM) (figure 2C).

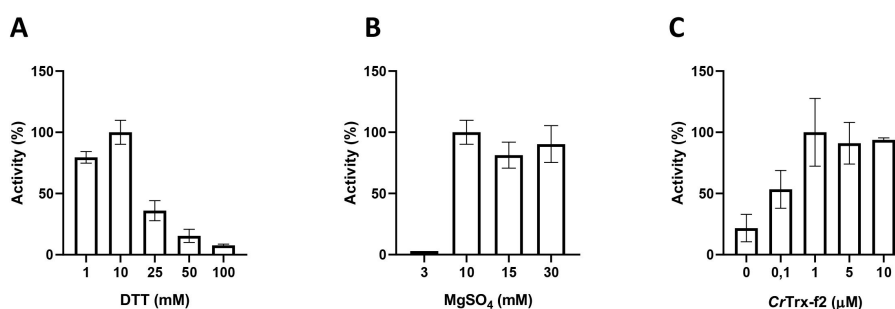


Figure 2.

Functional characterization of CrSBPase in vitro.

Reported enzymatic activity of CrSBPase assayed for (A) reduction by dithiothreitol (DTT), (B) Magnesium sulfate ($MgSO_4$), and (C) recombinant thioredoxin f2 from *Chlamydomonas reinhardtii* (CrTRX-f2).

Redox controlled dynamics of CrSBPase

CrSBPase crystallized under reducing treatment (10 mmol.L^{-1} TCEP). Among the eight subunits in the asymmetric unit, C115-C120 disulfide is absent in two chains (subunits C coloured in magenta and E coloured in salmon, figure 3D and 3F), while the other six subunits still have the C115-C120 disulfide bridge (figure 3B,C,E,G,H,I). All other cysteines are

unaffected by the reducing treatment. These six subunits present mildly variable conformations of the A₁₁₃SCAGTAC₁₂₀ lasso conformation similar to the untreated oxidized structure of CrSBPase (figure 3A). Reduction of the C115-C120 disulfide increases local disorder (supplementary figure 3), and the main chain electron density became uninterpretable over a few residues for subunits C (residues 115-118, 124-127), and E (residues 119-127).

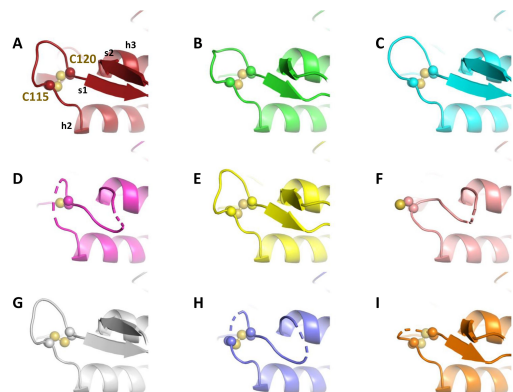


Figure 3.

Crystallographic structures of CrSBPase under reducing treatment.

(A-I) Aligned structures of CrSBPase protomers without redox treatment (A, 7b2o chain A) or in the presence of 10 mmol.L⁻¹ TCEP reducing agent (B-I, 7zuv chains A-H).

Molecular dynamics (MD) simulations of the reduced structure were performed by starting from a dimer of the crystallized oxidized form and forcing residues C115 and C120 to be reduced in a thiol state (-SH). We then ran two independent replicas of 20 μ s of MD. In addition, we also ran two independent replicas of 2 μ s of MD in the oxidized form. In one of the replicas of the reduced form (MD1), the protein structure barely changed as observed in the SG(C115)/SG(C120) distance that plateaued at ~ 7 Å, as well as in the RMSD with respect to the starting conformation that stayed around 2 Å (see supplementary figures 7-8 for plots and 12 for images of the structures). However, in the other replica of the reduced form (MD2), significant changes of conformation appeared: the SG(C115)/SG(C120) distance ended up at ~ 10 Å with peaks at 20 Å and the RMSD ended up at 3.7 and 2.8 Å (supplementary figures 7-8). The difference between MD1 and MD2 is stronger when one focuses on the RMSD of the mobile motif (residues 112 to 131, supplementary figure 9). We observed that after 17 μ s of MD all chains adopted an equilibrated conformation and then fluctuated around that conformation (supplementary figure 9); representative structures extracted from MD were thus obtained from a clustering on the time frame (or block) 17-20 μ s.

The two chains A and B from MD2 adopt a different final conformation (see figure 4A for the overlap with the crystallized oxidized structure and figure 4B for the overlap with the crystallized reduced structure), and we cannot directly conclude on which one is the most stable. Moreover, the residues 115 to 128 of each chain from MD1/2 do not overlap well with the crystallized reduced structure which means that neither of the two chains conformations from MD simulations has converged towards the crystallized state. However, we can observe in the reduced crystallized structure that residues 134-148 form an α -helix which is also present in chain B from MD2 whereas this structure is not kept in chain A from MD2 (see green chain on the right of figures 4A-B). In addition, mobile motif from chain B seems to open towards the solvent in a similar way with what was observed for FBPase in the reduced state. Thus, we conclude that chain B from MD2 is more representative of the true reduced conformation than chain A from MD2. If we compare the mobility of chain B from MD2 in the reduced state with one chain in the oxidized state (also chain B from MD2), we can see that the mobile motif is overall more flexible in the reduced state, which is especially true for residues around C120 (residues 118 to 123, supplementary figure 10). We then analyzed the RMSD of residues 112 to 131 from chain B during MD2 (supplementary figure 11): residues 112, 130 and 131 barely move which is expected since they belong or are directly connected to the two α -helices flanking the mobile motif. At ~ 6 μ s of the trajectory

almost all other residues (with the exception of C120) display an abrupt change of conformation, with a different magnitude though: starting from ~ 2 Å, the RMSD jumped to 3 Å for some residues and to 15 Å for other residues (supplementary figures 8, 9, and 11). At ~ 13.5 μ s, another change occurs which involves C120 but not all the other residues. Thus, we conclude that the opening of the motif happens in a concerted way and not sequentially residue per residue.

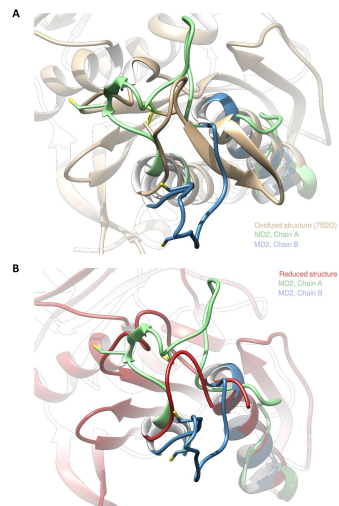


Figure 4.

Molecular dynamics simulation of CrSBPase after reduction.

(A) Overlap of the crystallographic structure of oxidized SBPase and representative structures of equilibrated reduced SBPase during molecular dynamics simulation 2 (MD2). For structures extracted from MD, only residues 109 to 148 are displayed since most of the other residues are closely overlapping those of the crystallographic structure. **(B)** Overlap of the crystallographic structure of reduced SBPase and representative structures of equilibrated reduced SBPase during MD2. For structures extracted from MD, only residues 109 to 148 are displayed.

Oligomeric states of CrSBPase

Size-exclusion profile of the purified native protein is polydisperse with at least three distinct species of CrSBPase (supplementary figure 1A,C). Notably, we observed that the protein elutes as a main peak of 79 kDa, close to the mass of a homodimer (theoretical mass from monomer sequence: 36,176.7 Da). This peak is broadened by a later-eluting shoulder, with the apparent molecular mass of the tailing species at 48 kDa suggesting the presence of a monomeric state of CrSBPase. The third species elutes at an apparent molecular mass of 145 kDa, that fits with a homotetramer. According to the absorbance of each peak and considering that they represent all states of purified and oxidized CrSBPase in solution, we estimate a repartition of species of 10:53:37 for the tetramer:dimer:monomer mixture (total = 100). Re-injection of the dimer peak fraction over size-exclusion chromatography yielded a similar mixed profile, supporting a dynamic equilibrium over the time of the experiment (>2 h) (data not shown).

CrSBPase was crystallized in space groups $P2_12_12$ or $P12_11$ with respectively 6 and 8 protomers in the asymmetric unit. Proteins Interfaces Structures and Assemblies (PISA) analysis reveals that protomers of both crystals pack as homodimers with extensive subunit interface of $4\,200\text{ Å}^2$ in a similar manner (figure 4A) (Krissinel and Henrick 2007). In the asymmetric unit of CrSBPase crystallized in reducing conditions (7ZUV) or within crystallographic neighbouring units of CrSBPase in oxidized state (7B20), CrSBPase further packs into a homotetramer, a dimer of dimers (figure 4B). The apparent molecular mass of SBPase extracted from *Chlamydomonas* cultures cultivated in TAP medium under $40\text{ }\mu\text{E.m}^{-2}.\text{s}^{-1}$ illumination is comprised between 130 and 40 kDa with the homodimer representing the predominant state as supported by immunoblot analysis (supplementary figure 4). To further investigate the oligomeric dynamics of CrSBPase, we performed small-angle X-ray scattering (SAXS) analysis of untreated or reduced enzyme *in vitro* which reveals an increase in the protein radius of gyration from 39 to 62 Å upon reduction (supplementary figure 5). This observation implies a change in the oligomeric state of CrSBPase correlating with the redox state. Besides, after a week at 4 °C, the purified protein partially precipitates

into an amorphous aggregate that treatment with 10 mmol.L⁻¹ TCEP turns into liquid-liquid phase separations (supplementary figure 6). Since the LBH redox module is positioned at the homodimer interface (figure 5C), we postulated that the reduction of the C115-C120 disulfide bond and the subsequent local conformation changes depicted by MD affect the oligomer equilibrium of the enzyme. As a matter of fact, we observed that protein mutants C115S or C120S, which are unable to form the C115-C120 disulfide bond, elute in a different oligomer ratio than wild-type CrSBPase (supplementary figure 1C). Indeed, we observed a relative accumulation of monomer and tetramer species at the expense of the dimer state in both mutants. This suggests that C115-C120 is involved in the stabilization of the dimer. Meanwhile, the activity of the two Cys mutants still responded to DTT/Mg-dependent activation resembling the response of the wild-type protein. No activity was detected for native and untreated forms, while specific activities of C115S and C120S mutants upon treatment with DTT and MgSO₄ were respectively 96% and 107% with respect to wild-type maximum activity (13.3 ± 2.5 and 14.9 ± 0.9 $\mu\text{mol}(\text{NADPH}) \text{ min}^{-1} \text{ mg}(\text{SBPase})^{-1}$ for C115S and C120S mutants, respectively). Based on these data, we can thus conclude that C115-C120 disulfide bond is not the sole thiol-dependent regulatory mechanism involved in the modulation of CrSBPase activity and that the redox dependency of CrSBPase catalysis likely depends on other cysteine pairs. Future studies are required to shed light on the identity of cysteine residues that contribute to CrSBPase catalytic modulation other than C115 and C120.

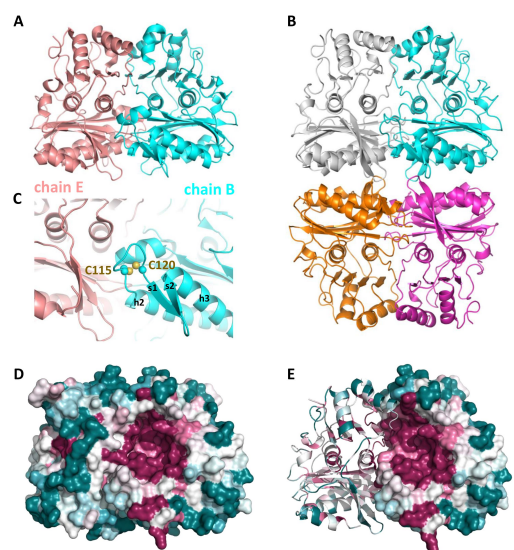


Figure 5.

Oligomeric structure of CrSBPase.

(A) Asymmetric unit dimer of untreated CrSBPase (7b2o). Chains are represented in cartoon and coloured cyan (chain B) and salmon (chain E). **(B)** Asymmetric unit homotetramer under reducing treatment (7zuv). Chains A (cyan), C (magenta), F (white), and H (orange) belong to the same asymmetric unit. **(C)** Close-up view on figure A homodimer interface. Loop 113-ASCAGTAC-120 from chain B (in cyan) is in 5-Å distance of neighbouring chain E (in salmon). C115 and C120 are bonded by a disulfide bridge. **(D)** Surface representation of 7b2o homodimer, coloured according to conservation score from teal (lowest conservation) to purple (highest conservation) among 150 aligned homologs.

Conclusions and perspectives

We report the first crystal structure of an SBPase from the microalga *Chlamydomonas reinhardtii*, a model for the molecular and cell biology of plants. SBPase is a photosynthetic enzyme involved in the regenerative phase of the CBBC cycle proved to be the target of multiple regulations that we map onto our structural model. The native folding of CrSBPase proved to be highly similar to that of SBPase from the moss *Physcomitrium patens* and to FBPase from chloroplast or cytoplasmic origins. Notwithstanding its propensity to form crystals, recombinant CrSBPase encompasses a range of oligomeric states, dominated by a homodimeric form susceptible to exchange for monomers, tetramers and higher-order assemblies. This oligomer equilibrium is dynamic and correlates with the presence of a disulfide bridge at the vicinity of the monomer-monomer interface, in the SBPase-specific LBH motif. We postulate that the reversible reduction/oxidation of the C115-C120 pair modulates the oligomerization of CrSBPase, through the induction of a local disorder that we

observed in the crystal structure of the protein under reducing conditions and that correlates with computations of molecular dynamics.

How this oligomeric exchange impacts the actual activity of the enzyme is an open question but should be aligned with previous reports on the allosteric character of the structurally similar FBPase enzyme which follows dimer/tetramer exchanges (Barciszewski, Wisniewski et al. 2016) with a Hill cooperativity coefficient close to 2 representing positive cooperativity in the context of a homodimer (Giudici-Orticoni, Buc et al. 1990). Lastly, systematic proteome mapping in the chloroplast of *Chlamydomonas* localized SBPase in a region surrounding the Rubisco pyrenoid nearby five other CBBC enzymes: phosphoglycerate kinase 1 (PGK), glyceraldehyde-3-phosphate dehydrogenases (GAP1, GAP3), fructose-bisphosphate aldolase (FBA3), ribulose-5-phosphate 3-epimerase (RPE1), and phosphoribulokinase (PRK) (Wang, Patena et al. 2022). Based on our structure-function study on recombinant SBPase, we can hypothesize that upon reductive activation SBPase is co-addressed to this liquid partition of the stroma where it contributes to CBBC reactions at optimized rates.

Table 1.

Crystallographic data collections and models building statistics.

	7b2o (oxidized)	7zuv (partially reduced)
Wavelength (Å)	0.9801	0.9801
Resolution range (Å)	46.62 - 3.095 (3.206 - 3.095)	48.54 - 3.11 (3.221 - 3.11)
Space group	P 2 ₁ 2 ₁ 2	P 1 2 ₁ 1
Unit cell (Å, °)	178.224 183.652 75.196 90 90 90	53.774 163.462 172.765 90 91.939 90
Total reflections	615155 (55989)	376091 (32376)
Unique reflections	45740 (4245)	53456 (5213)
Multiplicity	13.4 (12.9)	7.0 (6.2)
Completeness (%)	99.34 (94.20)	99.65 (97.17)
Mean I/sigma(I)	12.02 (1.72)	8.54 (1.30)
Wilson B-factor (Å ²)	64.82	68.45
R-merge	0.495 (1.877)	0.2492 (1.502)
R-meas	0.5144 (1.954)	0.2691 (1.64)
R-pim	0.1386 (0.5343)	0.1009 (0.6477)
CC1/2	0.983 (0.623)	0.988 (0.479)
CC*	0.996 (0.876)	0.997 (0.805)
Reflections used in refinement	45607 (4242)	53417 (5193)
Reflections used for R-free	1986 (183)	1983 (191)
R-work	0.1942 (0.3071)	0.1963 (0.3239)
R-free	0.2390 (0.3482)	0.2328 (0.3740)
CC(work)	0.944 (0.788)	0.956 (0.726)
CC(free)	0.912 (0.649)	0.949 (0.596)
Number of non-hydrogen atoms	14194	18714
Macromolecules	14190	18704
Solvent	4	10
Protein residues	1859	2451
RMS(bonds) (Å)	0.005	0.004
RMS(angles) (°)	0.75	0.73
Ramachandran favored (%)	94.26	94.80
Ramachandran allowed (%)	5.52	4.70
Ramachandran outliers (%)	0.22	0.50
Rotamer outliers (%)	0.13	0.55
Clashscore	4.88	6.60
Average B-factor (Å ²)	67.00	68.41
macromolecules (Å ²)	67.01	68.40
solvent (Å ²)	49.59	94.37

Statistics for the highest-resolution shell are shown in parentheses.

Supplementary information

Supplementary table 1.

Primers used for point mutagenesis.

5'-forward C115S	CCGCACCGCCTCGAGCGCCGGTACCGCCTGCGTG
3'-reverse C115S	CACGCAGGCGGTACCGGCGCTCGAGGCGGTGCGG
5'-forward C120S	GCGCCGGTACCGCCAGCGTGAACAGCTTCGGCG
3'-reverse C120S	CGCCGAAGCTGTTACGCTGGCGGTACCGGCGC

Supplementary figure 1. Purification of recombinant CrSBPase.

A. Size-exclusion chromatography profile recorded at absorbance $\lambda = 280$ nm on Superdex 200 26/600 GL column. B. Electrophoresis of affinity-chromatography fractions on 12% acrylamide gel in denaturing and reducing condition. Gel was stained with Coomassie blue. Mw: molecular mass standards ladder. C. Size-exclusion chromatograms of CrSBPase wild-type (blue), mutant C115S (orange), mutant C120S (gray).

Supplementary figure 2. SBPase multiple sequences alignment.

CrSBPase mature sequence was used for a BLAST search. Retrieved homolog sequences were aligned and coloured-coded according to residue conservation. Illustration made with EndScript (Robert and Gouet 2014).

Supplementary figure 3. Related to figure 3. Local disorder of the A₁₁₃SCAGTAC₁₂₀loop.

Main chain was traced according to crystallographic b-factor, with large orange sections representing high b-factor values and thin blue sections representing low b-factors. A-I. Aligned structures of CrSBPase protomers without redox treatment (A, 7b2o chain A) or in the presence of 10 mmol.L⁻¹ TCEP reducing agent (B-I, 7zuv chains A-H).

Supplementary figure 4. Size-exclusion fractionation of *Chlamydomonas* cell extracts.

A. *Chlamydomonas* cell culture was harvested, lysed and the soluble fraction of the lysate was loaded on Superose6 16/600 size-exclusion column. Chromatography fractions were analyzed by western blot with anti-CrSBPase primary antibodies. First membrane was loaded with fractions eluted from 40 to 80 mL. Second membrane was loaded with fractions eluted from 80 to 120 mL. M lane is loaded with molecular mass standards ladder. Recombinant CrSBPase was loaded on last lane.

Supplementary figure 5. Size-exclusion chromatography coupled to small angle X-rays scattering (SEC-SAXS) of CrSBPase.

A. X-ray scattering curve $\log(I) = f(s)$ for pure CrSBPase untreated (dark blue) or treated (light blue) with 10 mmol.L⁻¹ dithiothreitol (DTT). B. Radius of gyration of untreated protein was

computed by ATSAS Primus Guinier Wizard as $R_g = 39.14 \pm 0.06 \text{ \AA}$ (Franke, Petoukhov et al. 2017). C. Radius of gyration of DTT-treated protein was computed as $R_g = 62.47 \pm 0.38 \text{ \AA}$.

Supplementary figure 6. Phase separation of reduced CrSBPase precipitates.

CrSBPase at 2 mg/mL formed amorphous precipitates after storage at 4 °C for a week. The suspension was sedimented by centrifugation at 20,000 *g* for 20 minutes and a microliter drop of the supernatant or of the pellet were treated with 10 mmol.L⁻¹ reducing agent TCEP and visualized by binocular optical microscope. A. Untreated precipitate. B. TCEP-treated precipitate. C. Untreated supernatant. D. TCEP-treated supernatant.

Supplementary figures 7-11. Computational detail of molecular dynamic simulation.

Supplementary figure 7.

SG(Cys115)/SG(Cys120) distances along the MD trajectories for each chain (left: oxidized state, right: reduced state). Light colours: raw data saved every 100 ps, dark colours: moving average on 20ns blocs.

Supplementary figure 8.

RMSD of main chain along the MD trajectories for each chain (left: oxidized state, right: reduced state). Light colours: raw data saved every 100 ps, dark colours: moving average on 20 ns blocs.

Supplementary figure 9.

RMSD of main chain of the mobile motif in the reduced state (residues 112 to 131) along the MD trajectories for each chain (left: RMSD with respect to the starting conformation of MD, right: RMSD with respect to the conformation at 20 μ s). Light colours: raw data saved every 100 ps, dark colours: moving average on 20 ns blocs.

Supplementary figure 10.

RMSF per residue for each chain (top row: full sequence, bottom row: zoom on residues 110 to 170; left column: oxidized state, middle column: reduced state, right column: comparison of one chain from each state).

Supplementary figure 11.

RMSD of main chain of each residue from the mobile motif along the trajectory of chain B from MD2. Data are moving average on 200 ns blocs.

Supplementary figure 12. Structures of CrSBPase retrieved from molecular dynamics simulations 1 and 2.

Two independent replicas of 20 μ s were performed. In one of them (MD1), the overall structure of both chains of the enzyme barely changed. To illustrate this, we present in supplementary figure 12 the overlap between the crystallographic oxidized structure and representative structures extracted from MD (based on clustering the last 3 μ s since each chain does not display significative changes after 17 μ s). A. Overlap of the crystallographic structure of oxidized CrSBPase and representative structures of equilibrated reduced SBPase during MD1. For structures extracted from MD, only residues 111 to 132 are displayed since the other residues are closely overlapping those of the crystallographic structure. B. Overlap of the crystallographic structure of oxidized SBPase and representative structures of equilibrated reduced CrSBPase during MD2. For structures extracted from MD, only residues 109 to 148 are displayed since most of the other residues are closely overlapping those of the crystallographic structure.

Authors contributions

JH, NC: Conceptualization; JH, NC, MS, TLM: Data curation; JH, NC, MZ: Formal analysis; JH: Funding acquisition; JH, NC, MZ, SDL, MS, TLM: Investigation; JH, NC, MZ, MS, TLM: Methodology; JH, SDL: Project administration; JH, NC, SDL: Resources; NC: Software; JH, MZ: Supervision; JH, NC, MZ, SDL, MS, TLM: Writing - original draft; JH, NC, MZ, SDL, MS, TLM: Writing - review & editing.

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Conflicts of interest statement

The authors report no conflict of interest relevant to the conduct or report of this research.

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Reviewer #1 (Public Review):

In this study, Le Moigne and coworkers shed light on the structural details of the Sedoheptulose-1,7-Bisphosphatase (SBPase) from the green algae *Chlamydomonas reinhardtii*. The SBPase is part of the Calvin cycle and catalyzes the dephosphorylation of sedoheptulose-1,7-bisphosphate (SBP), which is a crucial step in the regeneration of ribulose-1,5-bisphosphate (RuBP), the substrate for Rubisco. The authors determine the crystal structure of the CrSBPase in an oxidized state. Based on this structure, potential active site residues and sites of post-translational modifications are identified. Furthermore, the authors determine the CrSBPase structure in a reduced state revealing the disruption of a disulfide bond in close proximity to the dimer interface. The authors then use molecular dynamics (MD) to gain insights into the redox-controlled dynamics of the CrSBPase and investigate the oligomerization of the protein using small-angle X-ray scattering (SAXS) and size-exclusion chromatography. Despite the difference in oligomerization, disruption of this

disulfide bond did not impact the activity of CrSBPase, suggesting additional thiol-dependent regulatory mechanisms modulating the activity of the CrSBPase.

The authors provide interesting new findings on a redox-mechanism that modulates the oligomeric behavior of the SBPase, however without investigating this potential mechanism in more detail. The conclusions of this manuscript are mostly supported by the data, but they should be more carefully evaluated in respect to what is known from other systems as e.g. the moss *Physcomitrella patens*. This is especially of interest, as SBPase was previously reported to be dimeric, whereas for FBPase a dimer/tetramer equilibrium has been observed.

1.) Given that PpSBPase has been already characterized in detail, the authors should provide a more rigorous comparison to the existing data on SBPases. This includes a more conclusive structural comparison but also the enzymatic assays should be compared to the findings from *P. patens*. Do the authors observe differences between the moss and the chlorophyte systems, maybe even in regard to the oligomerization of the SBPase?

2.) The authors should include the control experiments (untreated SBPase) and the assays performed with mutant versions of the SBPase, which are currently only mentioned in the text or not shown at all.

3.) The representation of the structure in figures (especially Figures 1 and 3) should be adjusted to match the author's statements. In Figure 1, the angle from which the structure is displayed changes over the entire figure making it difficult to follow especially as a non-structural biologist. Furthermore, important aspects of the structure mentioned in the text are not labeled and should be highlighted, by e.g. a close-up. Same holds true for Figure 3 that currently mostly shows redundant information.

4.) The authors state that mutation of C115 and C120 to serine destabilize the dimer formation, while more tetramer and monomer is formed. As the tetramer is essentially a dimer of dimers, the authors should elaborate how this might work mechanistically. In my opinion, dimer formation is a prerequisite for tetramer formation and the two mutations rather stabilize the tetramer instead of destabilizing the dimer.

Reviewer #2 (Public Review):

The central theme of the manuscript is to report on the structure of SBPase - an enzyme central to the photosynthetic Calvin-Benson-Bassham cycle. The authors claim that the structure is first of its kind from a chlorophyte *Chlamydomonas reinhardtii*, a model unicellular green microalga. The authors use a number of methods like protein expression, purification, enzymatic assays, SAXS, molecular dynamics simulations and xray crystallography to resolve a 3.09 Å crystal structure of the oxidized and partially reduced state. The results are supported by the claims made in the manuscript. One of the main weakness of the work is the lack of wider discussion presented in the manuscript. While the structure is the first from a chlorophyte, it is not unique. Several structures of SBPase are available. As the manuscript currently reads, the wider context of SBPase structures available and comparisons between them is missing from the manuscript. Another important point is that the reported structure of crSBPase is 0.453 Å away from the alphafold model. Though fleetingly mentioned in the methods section, it should be discussed to place it in the wider context.